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Business-centric data solutions for safeguarding animal agriculture in the United States of America

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Summary

Business-centric solutions to data-related problems often yield the greatest positive impacts and improvements for private enterprises but are challenging to design and implement at scale within government agencies. The core mission of the Veterinary Services of the United States Department of Agriculture (USDA) Animal Plant Health Inspection Service is to safeguard animal agriculture in the United States of America, and effective data management underpins these efforts. As this agency works to assist data-driven decision-making in animal health management, it continues to use a blend of best practices from Federal Data Strategy initiatives and the International Data Management Association framework. This paper describes three case studies that focus on strategies to improve animal health data collection, integration, reporting and governance for animal health authorities. These strategies have enhanced the way USDA's Veterinary Services execute their mission and core operational activities for prevention, detection and early response to support disease containment and control.

Keywords

Animal health – Automation – Business process improvement – Data collection – Data governance – Data management – Integration – Reporting – Standardisation.

Introduction

In 2019, the United States Federal Government Office of Management and Budget announced the creation of the first ever coordinated and integrated Federal Data Strategy (FDS). This began a ten-year initiative to provide a government-wide vision of how agencies should manage and use federal data. The FDS mission is to: 'fully leverage the value of federal data for mission, service, and the public good by guiding the Federal Government in practicing ethical governance, conscious design, and a learning culture' [1].

The FDS framework consists of 10 principles, 40 practices and 20 action steps to guide federal agencies on data management and use (Table I). The full framework is available online [1]. Although designed for senior executives in federal agencies, specifically Chief Data Officers, the principles and practices have broad and cross-cutting applications to teams, programmes and projects at all levels of a federal agency.

Table I

Federal data strategy principles

See [1] for detailed descriptions

Ethical governance	
1	Uphold ethics
2	Exercise responsibility
3	Promote transparency
Conscious design	
4	Ensure relevance
5	Harness existing data
6	Anticipate future uses
7	Demonstrate responsiveness
Learning culture	
8	Invest in learning
9	Develop data leaders
10	Practise accountability

The United States Department of Agriculture (USDA) is one of many federal agencies implementing the FDS framework.

Within the USDA, the Veterinary Services (VS) agency of the Animal and Plant Health Inspection Service (APHIS) has identified and created ways to apply these principles and practices to animal health data and information systems. Veterinary Services' mission is to improve the health, productivity and quality of life for animals and people, and to maintain and promote the safety and availability of animals, animal products and veterinary biologics [2].

The work of APHIS VS encompasses a vast number of activities, including: comprehensive surveillance for numerous animal commodity groups and disease programmes, laboratory information services, One Health, emergency preparedness, animal movements and traceability, veterinary accreditation and other emerging issues. To accomplish its programme goals, APHIS VS collaborates with local, state, tribal, national and international partners. To underpin many of these activities, APHIS VS leverages animal health data and information systems as strategic assets to deliver on its mission, serve the public and steward national livestock stakeholder resources.

Effective use of data requires a strong data management framework that provides detailed actions for the collection, storage, structure and maintenance of an organisation's data and information systems. Better data management leads to improved data quality and access; increased productivity, cost efficiency and data security; and informed decisions for operational planning [3]. The Data Management Association (DAMA) International has created a data management framework with ten core functional areas that organisations can use as a guide for data management strategy and implementation [3]. At the core of the framework is data governance, but it also includes security; architecture; database operations; document and content management; data development; reference, master and metadata management; data quality; integration and interoperability; and data warehousing and business intelligence [3].

The approach of APHIS VS to governing, managing and protecting data leverages both the principles and practices listed in the FDS and the core functional areas of the DAMA framework to promote efficient and appropriate data use and to optimise the value of animal health data to support programme activities. The data maturity process has been cyclical – establish a baseline, gain support, collaborate, and refine continuously. In operational terms, common FDS principles used by APHIS VS programmes include [1]:

- ensuring relevance (FDS Principle 4)
- harnessing existing data (FDS Principle 5)
- anticipating future uses (FDS Principle 6)
- demonstrating responsiveness (FDS Principle 7).

Collectively, these four principles represent conscious design, a process that creates a product that is responsive to the needs of the programme. By integrating FDS principles and practices with functional areas of DAMA data management, animal health programmes are better positioned to develop flexible data solutions that meet and adapt to business needs, rather than adjusting or limiting their business processes to conform to information technology (IT) systems.

This paper examines three case studies focused on animal health data needs to demonstrate how APHIS VS uses the DAMA framework and FDS principles to harness the value of existing data. These examples were selected because they resulted in data products and solutions that met the user requirements of animal health authorities, while supporting effective and efficient animal health management processes. Use of the DAMA framework and FDS principles, as demonstrated by these examples, may be helpful for other countries' animal health authorities as they take advantage of data to support regulatory and management decisions, while navigating the challenging space between information technology, data management and animal health.

Case studies

Improving data quality through the management and standardisation of knowledge

Avian influenza testing and reporting: an example

Within the United States of America (USA), State Animal Health Officials and poultry industry partners perform testing for avian influenza (AI) as part of a cooperative arrangement for AI surveillance, organised by APHIS VS. The cooperators can use the funding provided to pay for testing at any public laboratory of their choosing (e.g. a university laboratory, the National Animal Health Laboratory Network, a referral laboratory). This AI testing programme complements several other systems in the USA [4, 5, 6].

All testing funded by this programme must be reported to APHIS on a quarterly basis. Additionally, any positive AI test results, regardless of funding source, must be reported to APHIS VS immediately. While this system has been very successful at supporting claims of disease freedom, the process of collating and reporting these data has been extremely labour intensive. A recent project aimed to improve the efficiency and data quality of AI reporting by using data standards and developing a knowledge management strategy.

Data management not only sets the foundation for how IT systems share and integrate information but also supports efficient and effective collection and use of the information [6]. Understanding and defining business rules (i.e. business concepts and governance) is integral to data management. Business rules can be achieved through a knowledge management strategy and the development of a concept model.

Effective business rules often apply a vocabulary standard. The vocabulary standard must be more than a simple listing of terms; it must also include modelled relationships between terms to fully depict the underlying knowledge (e.g. 'each report is submitted by a contributor who is affiliated with a cooperator located in a state'). A concept model focuses on structuring business terms according to the connections between them. A strong vocabulary standard, supported by clear business language and definitions, along with an underlying concept model, helps to lay the groundwork for data quality. Together, they provide an integrated point of knowledge and ensure all business rules and other forms of communication (e.g. business policies, IT requirements) are consistently expressed.

For this surveillance programme, a modernisation effort replaced a spreadsheet-based reporting tool, which required time-consuming manual review and correction of invalid test results, with an online data entry system that included rules to enforce data quality. The development of this new system and a 'best practice' knowledge management strategy resulted in a set of vocabulary terms, definitions and business rules that served as the basis for the IT system and automated data quality support. In addition to providing meaning and coherency to business rules, processes and IT system requirements, the well-structured (modelled) vocabulary also expressed AI surveillance guidelines and policy in a robust yet approachable form for the animal health workforce.

The strategy to develop the modernised AI system used perspectives and guidance from books [7, 8], training resources from the Business Rules Solutions company (https://www.brsolutions.com/company) and the newsletter of the Business Rules Community online (https://www.brcommunity.com/index.php). For example, the modernised AI surveillance reporting tool carried out the recommendation to treat any aspect of operational business knowledge that might change over time as a business rule rather than a vocabulary term [7]. A business rule's purpose is to shape (govern) day-to-day business activity and prevent undesirable situations that could occur [7], whereas a vocabulary term is a shared concept, based on defining criteria [8], but it does not describe any actions that affect business activity.

The value of applying the recommendation for separating rules from terms became apparent as the project progressed and two types of VS stakeholders were identified and engaged [9]. The first type of stakeholder was responsible for supporting the system's operation, including the ownership of the collected data, and managed the cooperative agreement system and guidelines. This stakeholder group has a strong interest in developing business (governance) rules that maximise data accuracy and minimise the need for manual correction of data errors. The business rules collected from this group (e.g. the total number of pooled bird specimens allowed for testing with the USDA-licensed type A influenza antigen capture immunoassay – which may change over time as testing guidelines are updated) served as the basis for the IT data management system requirements [10] and contributed to improved data quality in the modernised tool.

The second type of stakeholder, responsible for ownership of the vocabulary, had the broadest knowledge and created the vocabulary terms (e.g. which traits were used to classify the different types of birds tested). This stakeholder group has a strong interest in developing accurate definitions and making them available to cooperators using the modernised reporting tool [11]. By engaging stakeholders directly and intentionally, the outcomes contribute to improved data quality, because cooperators are able to review and confirm their understanding of terms prior to data entry. Well-defined terms are crucial since the data may eventually be integrated with other avian testing data or systems.

Knowledge sharing between APHIS VS AI disease subject matter experts, stakeholders involved in the surveillance programme and IT team members can reduce the chance of discrepancies between surveillance policy and the services offered by information systems [12]. By focusing on a well-structured vocabulary and guidelines, APHIS VS stakeholders were able to ensure that essential business needs were accurately represented and accounted for in the modernised system.

Both the old and the modernised reporting tool require workers to manually enter quarterly data for up to roughly 20 cells, depending on what was tested by the cooperator. However, the modernised system supports online data entry and carries out automated quality control with the application of business rules (such as detecting an incorrect number of birds tested, relative to the number of tests performed for the quarter) (Figure 1). This saves time by eliminating the need for manual review and having to request corrections.

In addition, the vocabulary terms and definitions are made accessible in the modernised system (Figure 2). This also contributes to the quality of the data as it improves the chance that system users share the same understanding of terms. Moreover, the experiences gained from this knowledge-based approach instilled confidence in the data collected and reported by the information system for those stakeholders who require high-quality evidence to reach export trade agreements with foreign partners.

Enhancing surveillance by developing the reporting system

Equine infectious anaemia: an example

Equine infectious anaemia (EIA) is a potentially fatal, blood-borne disease caused by the EIA virus (EIAV) and often naturally transmitted by blood-feeding insect vectors [13].

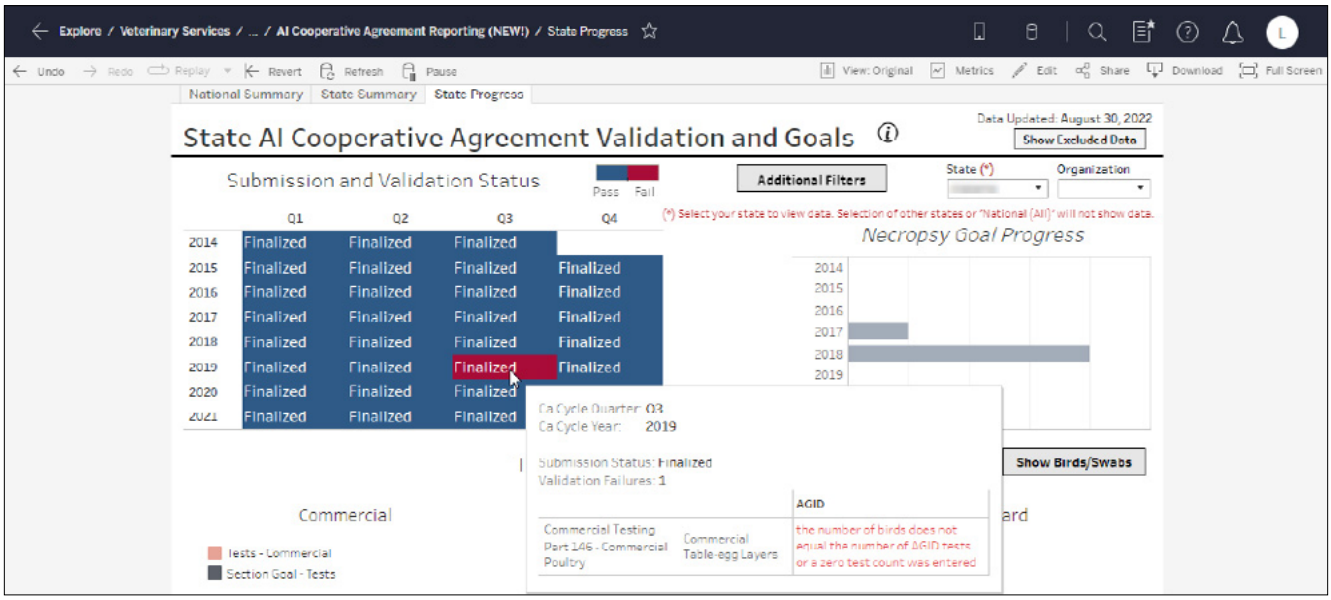

Figure 1

Avian influenza testing validation

Cooperators may visualise the quarters and years in which their submitted reports failed at least one business rule (red cell). If they hover over the red cells they will see the list of rules not successfully validated (depending on type of bird or facility sampled and type of testing performed). System access parameters are established by the administrator to allow cooperators to view only the data for their own state

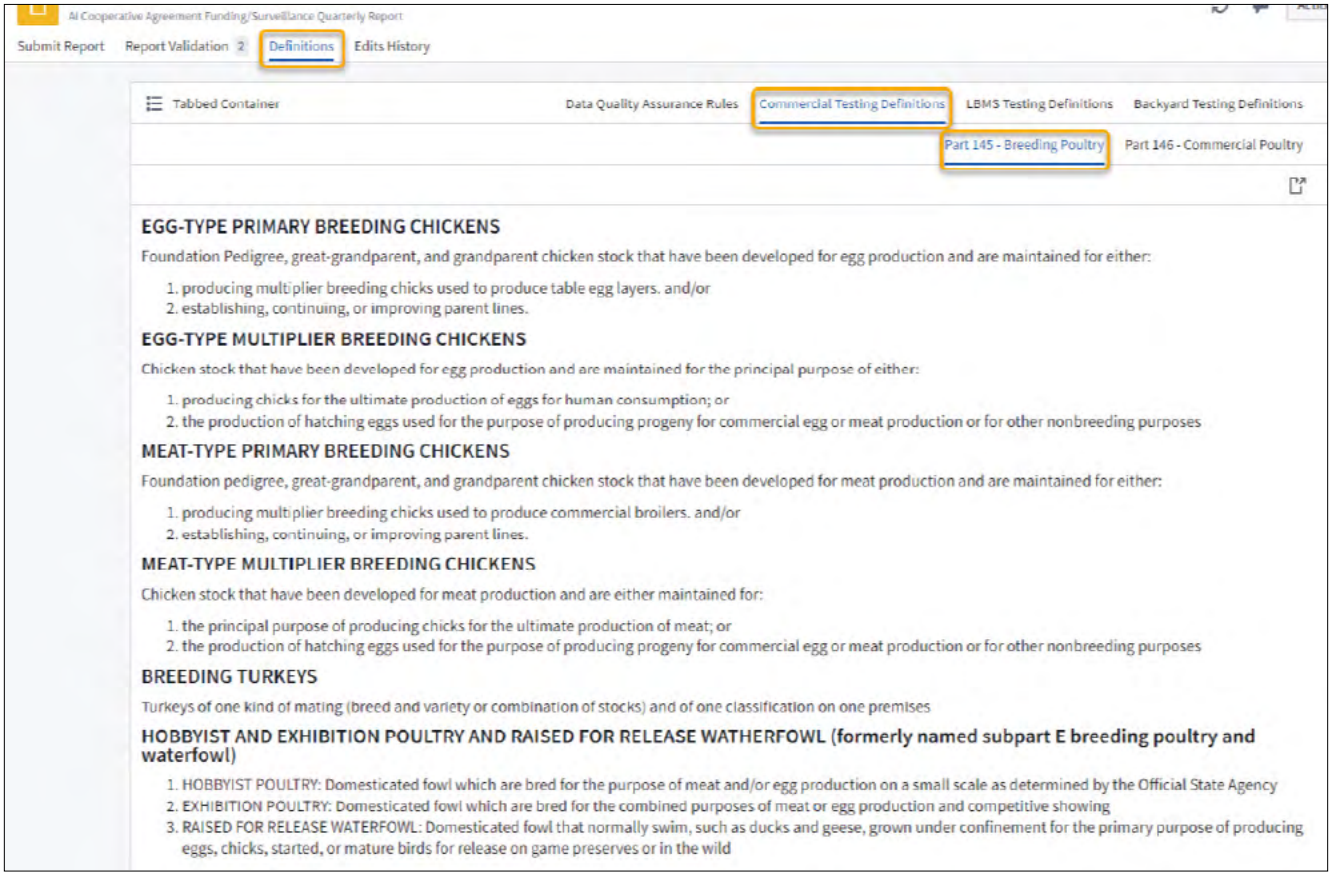

Figure 2

Avian influenza testing vocabulary terms and definitions

The modernised system makes vocabulary terms and definitions accessible to cooperators to create a common understanding of the precise types of birds and facilities sampled

It affects members of the Equidae family (horses and related equids), is present in many countries and creates a significant economic burden [14]. Initial exposure to EIAV causes an acute stage of infection with severity ranging from mild fever to sudden death [13]. If the animal survives, chronic and inapparent stages of infection follow, and infected equids become lifelong contagious carriers of EIAV [13]. No vaccine or effective treatment is currently available for EIA, meaning that the only disease management options are permanent isolation or euthanasia. As a result, preventing infection is essential for the overall control of EIA and protection of the equine industry [13]. During the past five decades of EIA control in the USA, surveillance, widespread testing and management of infected horses have proven to be the best methods for preventing EIA transmission [14, 15].

Surveillance and testing are key components of EIA control, especially given the variable clinical severity of EIA [15]. In 2019, APHIS VS released new guidance for EIA testing efforts [16], which built upon previous successes and aimed to further reduce EIA prevalence by enhancing oversight of EIA testing. Equine infectious anaemia testing is restricted to laboratories specifically trained and approved by the USDA National Veterinary Services Laboratories (NVSL), and the new guidance enhanced inspection, documentation and reporting standards. An important component of this guidance requires over 400 approved EIA testing laboratories to report monthly totals of EIA tests and summary results to the APHIS VS equine health team, via monthly reports [16].

To facilitate data collection, integration and reporting of the large number of EIA reports required each month as part of the new guidance standards, APHIS VS developed a new laboratory-reporting pipeline. During initial programme development, the existing NVSL master list of EIA-approved laboratories and contact information was used to open communication with the laboratories. The equine health team liaised with the EIA testing laboratories to set expectations and the ground rules for reporting. Early on, the team recognised that EIA testing laboratories have a diversity of technology for data handling and a wide range of IT skills among their staff. For this reason, the laboratories needed general and widely available tools to create and submit their EIA monthly reports to APHIS VS. The laboratory-reporting pipeline was developed to respond to these requirements and included:

- Microsoft Excel and email as the primary tools for creating and submitting reports
- programmed steps to access, curate and process the reports
- natural language processing (NLP) algorithms to achieve data standardisation
- data-marts (subject-oriented databases for specific users) for internal reporting and analysis of the received data [17, 18].

Limited time was available, given that USDA policy requiring monthly summary reporting from the laboratories became effective almost immediately. In lieu of a complex information system, a highly customised, lightweight Excel workbook was created as a template for EIA monthly reports, which included data validation rules, reference data and user hints to expedite and assist accurate reporting [19]. Laboratories complete monthly reports of aggregate-level data (e.g. the total number of tests and total number of positives) and send them to an APHIS VS equine health email inbox, where the reports are automatically downloaded and the data are cleaned, combined into a single data set and loaded onto an APHIS VS data system. The data management process includes automatically omitting duplicate reports and flagging files with errors for review. Laboratories with clearance to access APHIS VS systems also have the option of directly entering their monthly reports into an APHIS VS data system. The integrated data pipeline automatically combines the results from both reporting options into a single data set, and cleans and processes the data to ensure accuracy on summary reports. For example, one data management step automatically standardises laboratory contact information, using an NLP algorithm (term frequency–inverse document frequency) by comparing data from the reports against the NVSL-maintained master list of EIA testing laboratories [20, 21]. Finally, the prepared data are visualised and analysed across various metrics, including testing numbers, test types (enzyme-linked immunosorbent assay [ELISA] versus agar gel immunodiffusion [AGID]) and test results (positive, negative). The summary results are then published for both internal and public audiences (https://www.aphis.usda.gov/aphis/ourfocus/animalhealth/animal-disease-information/equine/eia/equine-infectious-anemia).

The laboratory-reporting pipeline has been successfully operating since 2019, shortly after the updated guidance was released [16]. This pipeline provided two key lessons in data quality. First, Microsoft Excel workbooks are desirable for their general familiarity and flexibility, but this is a disadvantage for reporting, which requires standard, consistent data formatting. Second, receiving reports via email was fast to develop, but also created challenges when trying to trace faulty reports back to the original submitter. To address these two broad issues, strong relationships and good communication with laboratory directors and staff were essential.

Likewise, the implementation of automated data management procedures has proven invaluable. In future, it is planned to shift all laboratories to entering their data directly into the APHIS VS system, using a simple online data entry form currently in use by the small number of laboratories who already have access to APHIS VS systems. This transition will eliminate the two data quality issues described above.

Even in its current format, this new pipeline receives several million test result reports for EIA surveillance annually,

successfully contributing to EIA control efforts to protect the equine industry (Figure 3).

Building efficiencies using digital tools

Approval of livestock markets: an example

Livestock transportation and concentration points, such as livestock markets and slaughter plants, handle large numbers of animals as they move within these centralised and structured places. Federal oversight and accurate tracing of animal movements at these locations are essential for disease traceability.

To qualify and operate as a federally approved livestock market, the legal operator of the facility must sign and execute an agreement and comply with the United States Code of Federal Regulations [22]. This agreement safeguards the livestock and facility operations, by ensuring that the livestock market complies with federal regulations on record keeping, identification, cleaning and disinfecting, maintenance of the facility and equipment, and livestock handling. It also ensures access to conduct necessary activities for disease detection and traceback, as well as facility cooperation and notification of livestock that are known or suspected to be infected with any infectious, contagious or communicable disease.

The APHIS VS Animal Disease Traceability (ADT) programme is dedicated to implementing a comprehensive animal disease traceability system to help identify and locate animals and trace their movements in the event of an outbreak. The ADT programme also supports authorities to ensure that interstate animal movements comply with federal regulations [22, 23, 24, 25, 26, 27, 28, 29, 30], and continuously develops and provides new tools to make traceability activities more efficient and effective. Recently, the ADT programme began an initiative to modernise livestock market agreements by using a more streamlined electronic approach.

Before 2021, the established process for approving livestock market inspections and agreements used paper forms and wet ink signatures transmitted by the United States Postal Service or by emailing the scanned documents after signature. An APHIS field officer would visit a facility, review the agreement with the facility owner/operator and retain an ink-signed copy of the agreement. The APHIS field officer would then send the agreement to the veterinarian in charge of the respective area, who would send the agreement to the State Animal Health Official, and subsequently to the APHIS Administrator (or an official designee) for final approval and signature. The APHIS ADT programme retained the paper copies, maintained the list of approved markets in the USA and shared the list on a public website.

This was a time-consuming process and depended on many manual steps for each agreement, including mailing, emailing or hand delivering copies of agreements to the required approving officials. After final approval, the records were manually stored and updated as necessary, using paper files and spreadsheets for over 1,200 federally approved livestock markets in the USA.

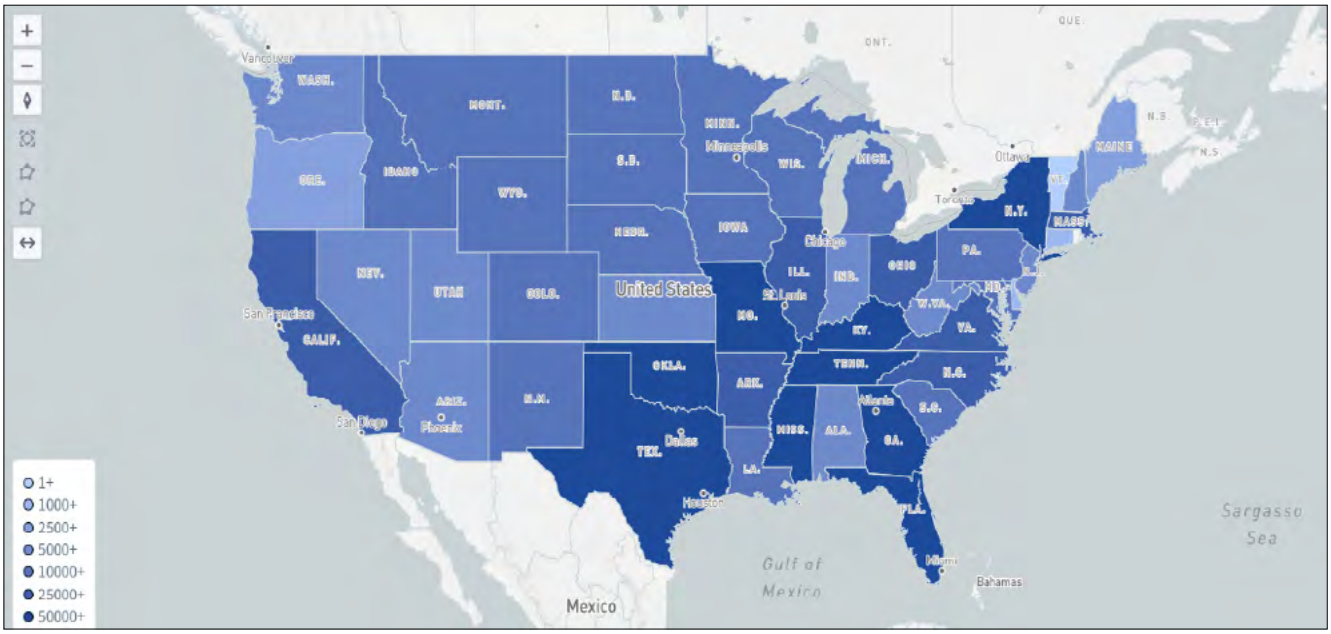

Figure 3

Geographic distribution of the number of equine infectious anaemia tests performed by laboratories in each state, in 2021

Map internally available to Animal and Plant Health Inspection Service (APHIS) Veterinary Services (VS) and State Animal Health Officials as a reporting output within the equine infectious anaemia (EIA) module of APHIS VS's Data Integration Services system. A total of 1,391,792 tests were reported by laboratories through the new EIA data pipeline in the 2021 calendar year

The main goal of the modernisation project was to introduce and adapt new technology and systems to improve the efficiency, effectiveness and accuracy of the livestock market agreement process. The new process updates the former process in three important ways.

a) It uses a fully electronic form.
b) It automates data curation and collation.
c) It includes an interactive real-time dashboard (map) on a public website.

The electronic form gives users the option to enter data while online or offline on smartphones, tablets or laptop computers. It improves the value of the information gathered by incorporating data quality checks and assisted form completion at the time of data entry, removing the possibility of erroneous or missing fields that need to get sent back for completion. It also facilitates and speeds data entry by providing the ability to select a market facility from a standard list of premises (reference data) and pre-populating the corresponding fields. In addition, the electronic form streamlines the signature routing workflow by sending email notifications to the approving officials, who can then log in and sign the form digitally. The form now stores data in a tabular electronic format, which enables the automation of data management and reporting processes. Scripts and routines clean and collate the data, summarise the results and update the list of approved markets daily. Facility locations are geocoded to create a map and table, with pinpoints of approved livestock markets on an interactive dashboard available on a public website (Figure 4, https://www.aphis.usda.gov/aphis/ourfocus/animalhealth/sa_livestock_markets). With electronic data, agreement records can be easily managed and accessed via an electronic library/repository of signed agreements. Animal Disease Traceability programme staff can readily query and analyse the approved agreement information and have additional metrics to assist with programme logistics and operations. For example, a separate internal dashboard was created for APHIS personnel to track the signature routing and completion of the agreements. In addition, staff can review the dates of a market's inspections to determine when another inspection should be scheduled in the future.

To achieve the change from paper to electronic records and maintain an accurate list of all approved markets, the ADT programme modified its policy and set a firm timeline for switching to this new electronic form and approval process. Following this switch, the average time to completion for an agreement has been reduced from weeks/months to days/hours. Not only has this project improved the inspection process, it also gives APHIS VS more real-time control and oversight of the markets to help with disease control and prevention.

Figure 4

Federally approved livestock markets

Interactive map of all federally approved livestock markets in the United States of America, available at https://www.aphis.usda.gov/aphis/ourfocus/animalhealth/sa_livestock_markets

Discussion

The three case studies presented here illustrate several activities where APHIS VS programme needs and processes did not fit into an existing information system. Each example varies greatly in terms of problem area, environment and diversity of stakeholders; yet the resulting solutions all used a business-centric approach, which followed best practices for business analysis (https://www.iiba.org/about-iiba), applied FDS principles and practices, and used core functions of the DAMA data management framework. By employing this business-centric approach to knowledge management, system development and process improvement, APHIS VS programmes developed unique and flexible data solutions to meet and adapt to business needs, rather than adjusting business processes to fit their IT systems.

Each case study highlights the application of several suggested FDS practices [1], together with the DAMA data management framework [3], to help animal health programmes maximise the value of their data.

The first case study applied knowledge management techniques, including mutually agreed upon vocabulary, definitions and rules, to simplify and improve AI surveillance reporting. Key FDS practices evident in this case study were leveraging data standards (No. 20), designing data for use and reuse (No. 29) and supporting federal and non-federal stakeholders (Nos 39 and 40).

The second case study described the development of a new EIA monthly laboratory-reporting pipeline designed to accept data from a large, diverse group of laboratories and automatically prepare the information for stakeholders. Crucial FDS practices applied in this case study were to assess and balance the needs of stakeholders (No. 2), identify opportunities to overcome obstacles in resourcing (No. 22) and leverage partnerships (No. 36).

The final case study introduced new technologies into an existing livestock market agreements approval process, automating the reporting procedure and greatly reducing the time needed to process agreements. Two essential FDS practices introduced here were to increase the capacity for data management and analysis (No. 27) and align data quality with intended use (No. 28).

Collectively, these solutions strengthened APHIS VS's ability to effectively and efficiently complete core operational activities. These practices and this framework may be helpful for other animal health authorities as they consider their own data management challenges.

Like many other countries, USDA APHIS VS is cultivating a business-centric approach to continuous improvement of data management and use. This ensures that animal health programme needs are met by developing data products and solutions designed to meet user requirements and support business processes, rather than adjusting business processes to use existing systems. Inherent in this approach is the use of conscious design when building data solutions, guided by four FDS principles [1]:

a) ensure relevance
b) harness existing data
c) anticipate future uses
d) demonstrate responsiveness [1].

These four principles enable the designer to maximise the value of existing data. In this way, APHIS VS is better positioned to create efficient and effective solutions that integrate with an existing portfolio of information systems and enhance the ability of VS to achieve programme goals.

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Des solutions de gestion des données centrées sur l'activité pour préserver les productions animales états-uniennes

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Résumé

S'agissant des problèmes en lien avec les données, les solutions centrées sur l'activité sont souvent celles qui génèrent le plus d'effets positifs et d'améliorations pour les entreprises du secteur privé, mais elles sont difficiles à concevoir et à mettre en œuvre à grande échelle au sein des agences gouvernementales. Les Services vétérinaires du Service d'inspection de la santé animale et végétale du département américain de l'Agriculture (USDA) ont pour mission centrale de préserver les productions animales états-uniennes ; une gestion efficace des données vient soutenir cet effort. Dans leur action d'appui aux processus décisionnels de gestion de la santé animale fondés sur les données, ces Services recourent à une combinaison de bonnes pratiques mises en œuvre aussi bien par les initiatives de la Stratégie fédérale sur les données que dans le cadre de l'Association internationale de gestion des données. Les auteurs décrivent trois études de cas sur des stratégies visant à améliorer la collecte, l'intégration, la notification et la gouvernance des données de santé animale afin de répondre aux besoins des autorités compétentes dans ce domaine. Ces stratégies ont permis aux Services vétérinaires de l'USDA de mieux s'acquitter de leur mission et d'améliorer leurs activités opérationnelles de prévention, de détection et de réaction rapide afin d'endiguer et contrôler les maladies.

Mots-clés

Amélioration des processus opérationnels – Automatisation – Collecte de données – Gestion des données – Gouvernance des données – Intégration – Notification – Santé animale – Standardisation.

Soluciones empresariales de gestión de datos para salvaguardar la producción animal estadounidense

L. Estberg, J. Luxton, K. Spiegel, A. Pelzel-McCluskey, B.L. Gomez & J.L. Vanden Eng

Resumen

Las soluciones eminentemente empresariales a problemas relacionados con los datos deparan con frecuencia los mejores frutos y resultados a la empresa privada, pero son difíciles de diseñar y aplicar a escala dentro de las administraciones públicas. Los Servicios Veterinarios adscritos al Servicio de Inspección Sanitaria de Animales y Plantas del Departamento de Agricultura de los Estados Unidos (USDA) tienen por principal cometido salvaguardar la producción animal estadounidense, labor que pasa en parte por una eficaz gestión de los datos. En su función de apoyo a la adopción de decisiones de gestión zoosanitaria basadas en los datos, este organismo sigue empleando una combinación de prácticas óptimas tomadas de iniciativas de la Estrategia Federal de Datos y de las pautas marcadas por la Asociación Internacional de Gestión de Datos. Los autores presentan y analizan tres ejemplos de estrategias para mejorar la obtención, integración, notificación y administración de datos zoosanitarios para las autoridades del ramo. Estas estrategias han conferido mayor eficacia a los Servicios Veterinarios del USDA en el cumplimiento de su misión y en la ejecución de sus principales actividades operativas de prevención, detección y pronta respuesta para ayudar a contener y combatir enfermedades.

Palabras clave

Administración de datos – Automatización – Gestión de datos – Integración – Mejora de los procesos operativos – Normalización – Notificación – Recogida de datos – Sanidad animal.

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